

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think>

To find the time it takes to download the file , I need to first calculate the time it takes to download the file without the interruption . Carla can download 2 GB / minute . The total size of the file is 200 GB .

So , the time it takes to download the file without the interruption is $200 \text{ GB} / 2 \text{ GB / minute} = 100$ minutes .

Now , I need to consider the interruption . The interruption occurs 40 % of the way through the download , which means Carla has downloaded 40 % of the file before the interruption . 40 % of 200 GB is $0.4 * 200 = 80 \text{ GB}$.

The time it takes to download 80 GB at a rate of 2 GB / minute is $80 \text{ GB} / 2 \text{ GB / minute} = 40$ minutes .

After the interruption , Carla has to restart the download from the beginning . So , she has to download the entire 200 GB again . As calculated earlier , this takes 100 minutes .

The total time it takes to download the file is the sum of the time before the interruption (40 minutes), the time for the interruption (20 minutes), and the time after the interruption (100 minutes).

Total time = $40 + 20 + 100 = 160$ minutes .

</think>< answer> 160 </answer>

